

leading coefficient of  $\Phi$  being  $\epsilon$  in place of 1, which introduces the factor  $\epsilon^i$ , we have

$$\epsilon^i \Psi_{m-i,0} = (-)^{(i-1)\frac{i}{2}} B_i',$$

where  $B_i'$  is the Bezoutian secondary of the  $(m-i-1)$ th degree in  $x$  to  $f$  and  $\phi$ ; but  $B_i'$  has been proved  $= B_i$ , the Bezoutian secondary of the same degree to  $f$  and  $\phi$ ; hence  $\Psi_{m-i,0} = (-)^{i\frac{i-1}{2}} B_i$ .

Art. 29. If now we return to the syzygetic equation,  $\tau f - t\phi + \mathfrak{S} = 0$ ,  $\mathfrak{S}$  may be treated as known, having in fact been completely determined as a function of the roots, as well in its most general form, as also so as to represent the simplified residues to  $f$  and  $\phi$  in the preceding articles; it remains to determine the values of  $\tau$  and  $t$  as functions of the roots corresponding to any allowable form of  $\mathfrak{S}$ , but I shall confine the investigation to the case where  $\mathfrak{S}$  is the lowest-weighted conjunctive or, which is the same thing, a simplified residue to  $f$  and  $\phi$  of any given degree in  $x$ ; each value of  $\frac{\tau}{t}$  will then represent one of the convergents to  $\frac{\phi}{f}$  when expanded under the form of a continued fraction. If  $\mathfrak{S}$  be of the  $\iota$ th degree in  $x$ ,  $\tau$  is of the degree  $(n-\iota-1)$  and  $t$  of the degree  $(m-\iota-1)$ . This being supposed, and calling  $n-\iota-1 = \nu$ ,  $m-\iota-1 = \mu$ , I say that  $t$  will be represented by  $G$  and  $\tau$  by  $\Gamma$ , where

$$G = (-)^{\iota} \Sigma (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_{\mu}}) \frac{\begin{bmatrix} h_{q_1} & h_{q_2} & \dots & h_{q_{\mu}} \\ \eta_1 & \eta_2 & \dots & \eta_n \end{bmatrix}}{\begin{bmatrix} h_{q_1} & h_{q_2} & \dots & h_{q_{\mu}} \\ h_{q_{\mu+1}} & h_{q_{\mu+2}} & \dots & h_{q_m} \end{bmatrix}},$$

and  $\tau$  is an analogous form  $\Gamma$ ;  $h_1, h_2 \dots h_m$ , as heretofore, being the roots of  $f$ , and  $\eta_1, \eta_2 \dots \eta_n$  of  $\phi$ . To fix the ideas and make the demonstration more immediately seizable, give  $m$  and  $n$  specific values; thus let  $m=5$ ,  $n=4$ ,  $\iota=2$ , so that  $\mu=5-2-1=2$ . Put  $\mathfrak{S}$  under the form  $\mathfrak{S}_{\iota,0}$ , so that  $\mathfrak{S}$  in the case before us

$$= \Sigma (x - h_{q_1})(x - h_{q_2}) \frac{\begin{bmatrix} h_{q_3} h_{q_4} h_{q_5} \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_{q_3} h_{q_4} h_{q_5} \\ h_{q_1} h_{q_2} \end{bmatrix}}.$$

Now make  $x = h_1$ , then  $f = 0$ , and  $\mathfrak{S}$  becomes

$$\Sigma (h_1 - h_{q_1})(h_1 - h_{q_2}) \frac{\begin{bmatrix} h_{q_3} h_{q_4} h_{q_5} \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_{q_3} h_{q_4} h_{q_5} \\ h_{q_1} h_{q_2} \end{bmatrix}},$$

that is

$$\Sigma \frac{\begin{bmatrix} h_1 \\ h_4 h_5 \end{bmatrix} \begin{bmatrix} h_1 h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_1 h_2 h_3 \\ h_4 h_5 \end{bmatrix}},$$

$h_1$  being kept constant in the above sum, but  $h_2, h_3, h_4, h_5$  being partitionable in all the six possible ways into two groups, as into  $h_4, h_5$ ;  $h_2, h_3$  in the term above expressed. This sum is evidently identical with

$$\Sigma \frac{\begin{bmatrix} h_1 h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_4 h_5 \end{bmatrix}}, \text{ that is } \begin{bmatrix} h_1 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix} \times \Sigma \frac{\begin{bmatrix} h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_4 h_5 \end{bmatrix}}.$$

Again,  $\phi$  becomes

$$\begin{bmatrix} h_1 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}.$$

Hence  $t = \frac{\mathfrak{S}}{\phi}$  becomes

$$\Sigma \frac{\begin{bmatrix} h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_4 h_5 \end{bmatrix}}.$$

But, when  $x = h_1$ ,  $\frac{G}{(-)^i}$  becomes

$$\begin{bmatrix} h_1 \\ h_2 h_3 \end{bmatrix} \frac{\begin{bmatrix} h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_1 h_4 h_5 \end{bmatrix}},$$

that is

$$= \frac{\begin{bmatrix} h_1 \\ h_2 h_3 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_1 \end{bmatrix}} \frac{\begin{bmatrix} h_2 h_3 \\ \eta_1 \eta_2 \eta_3 \eta_4 \end{bmatrix}}{\begin{bmatrix} h_2 h_3 \\ h_4 h_5 \end{bmatrix}},$$

$$= (-1)^i t.$$

Thus when  $x = h_1$ ,  $t = G$ . In like manner, when  $x = h_2$ , or  $h_3$ , or  $h_4$ , or  $h_5$ ,  $t$  always =  $G$ ; but  $t$  and  $G$  are both functions of  $x$  of the same, namely of only two, dimensions in  $x$ . Hence  $t$  is identical with  $G$ . So in general it may be proved, that whenever  $x = h_1$  or  $h_2$  or  $h_3 \dots$  or  $h_n$ ,  $t$  and  $G$ , which are each of only  $(m-1-i)$  dimensions in  $x$ , are equal. Hence universally  $t = G$ , as was to be shown. To find  $\tau$  we must avail ourselves of the symphic, or as we may better say (it being at the opposite extremity of the scale of forms), the antimorphic, value of  $\mathfrak{S}$  represented by  $\mathfrak{S}_0$ , taking care to preserve  $\mathfrak{S}$  strictly identical under both forms of representation, in point of sign as well as quantity. That is to say, we must make

$$\begin{aligned} \mathfrak{S}_{0,\iota} &= (-)^{\iota(m-\iota)} \Sigma (x - \eta_{q_1})(x - \eta_{q_2}) \dots (x - \eta_{q_\iota}) \frac{\begin{bmatrix} h_1, & h_2 & \dots & h_m \\ \eta_{q_{\iota+1}}, & \eta_{q_{\iota+2}} & \dots & \eta_{q_n} \end{bmatrix}}{\begin{bmatrix} \eta_{q_{\iota+1}}, & \eta_{q_{\iota+2}} & \dots & \eta_{q_n} \\ \eta_{q_1}, & \eta_{q_2} & \dots & \eta_{q_\iota} \end{bmatrix}} \\ &= (-)^\omega \Sigma (x - \eta_{q_1})(x - \eta_{q_2}) \dots (x - \eta_{q_\iota}) \frac{\begin{bmatrix} \eta_{q_{\iota+1}}, & \eta_{q_{\iota+2}} & \dots & \eta_{q_n} \\ h_1, & h_2 & \dots & h_m \end{bmatrix}}{\begin{bmatrix} \eta_{q_{\iota+1}}, & \eta_{q_{\iota+2}} & \dots & \eta_{q_n} \\ \eta_{q_1}, & \eta_{q_2} & \dots & \eta_{q_\iota} \end{bmatrix}}, \end{aligned}$$

where

$$\omega = \iota(m - \iota) + m(n - \iota),$$

so that

$$(-)^\omega = (-)^{m\iota - \iota + mn - m\iota} = (-)^{mn - \iota};$$

and consequently the same reasoning as was applied to  $t$  to prove  $t = G$ , will serve to show that  $-\tau = \Gamma$ , where

$$\Gamma = (-)^{mn} \Sigma (x - \eta_{\xi_1})(x - \eta_{\xi_2}) \dots (x - \eta_{\xi_\nu}) \frac{\begin{bmatrix} \eta_{\xi_1}, & \eta_{\xi_2} & \dots & \eta_{\xi_\nu} \\ h_1, & h_2 & \dots & h_m \end{bmatrix}}{\begin{bmatrix} \eta_{\xi_1}, & \eta_{\xi_2} & \dots & \eta_{\xi_\nu} \\ \eta_{\xi_{\nu+1}}, & \eta_{\xi_{\nu+2}} & \dots & \eta_{\xi_n} \end{bmatrix}},$$

or

$$\tau = (-)^\omega \Sigma (x - \eta_{\xi_1})(x - \eta_{\xi_2}) \dots (x - \eta_{\xi_\nu}) \frac{\begin{bmatrix} h_1, & h_2 & \dots & h_m \\ \eta_{\xi_1}, & \eta_{\xi_2} & \dots & \eta_{\xi_\nu} \end{bmatrix}}{\begin{bmatrix} \eta_{\xi_1}, & \eta_{\xi_2} & \dots & \eta_{\xi_\nu} \\ \eta_{\xi_{\nu+1}}, & \eta_{\xi_{\nu+2}} & \dots & \eta_{\xi_n} \end{bmatrix}},$$

where

$$\begin{aligned} \omega &= mn - 1 - m\nu = mn - 1 - m(n - \iota - 1) \\ &= m\iota + m - 1. \end{aligned}$$

Art. 30. I have not succeeded in throwing  $t$  and  $\tau$  under any other than the single forms for each above given, and it is remarkable that whilst apparently  $t$  and  $\tau$  admit only of this single representation,  $\mathfrak{S}$  admits of the variety of forms included under the general symbol  $\mathfrak{S}_{v,\iota-v}$  for a given value of  $\iota$ ; and it ought to be remarked that these forms, although the most perfectly symmetrical and exactly balanced representations, and for that reason possibly the most commodious for the ascertainment of the allotropic factor belonging to them respectively, by no means exhaust the almost infinite variety of modes by which the simplified residues, that is, the hekestobarytic, or if we like so to call them, the prime conjunctives, admit of being represented as functions of the roots of the given functions; for if in Art. 16, instead of writing

$$R = \frac{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots & h_{q_v} \\ \eta_{\xi_1}, & \eta_{\xi_2} & \dots & \eta_{\xi_\nu} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots & h_{q_v} \\ h_{q_{v+1}}, & h_{q_{v+2}} & \dots & h_{q_m} \end{bmatrix}} \times \frac{\begin{bmatrix} \eta_{\xi_1}, & \eta_{\xi_2} & \dots & \eta_{\xi_\nu} \\ \eta_{\xi_{\nu+1}}, & \eta_{\xi_{\nu+2}} & \dots & \eta_{\xi_n} \end{bmatrix}},$$

we had made

$$R = \frac{P(h_{q_1}, h_{q_2} \dots h_{q_v}; \eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_\nu})}{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots & h_{q_v} \\ h_{q_{v+1}}, & h_{q_{v+2}} & \dots & h_{q_m} \end{bmatrix} \times \begin{bmatrix} \eta_{\xi_1}, & \eta_{\xi_2} & \dots & \eta_{\xi_\nu} \\ \eta_{\xi_{\nu+1}}, & \eta_{\xi_{\nu+2}} & \dots & \eta_{\xi_m} \end{bmatrix}},$$

where  $P$  represents any function symmetrical in respect of  $h_{q_1}, h_{q_2} \dots h_{q_v}$ , and also in respect of  $\eta_{\xi_1}, \eta_{\xi_2} \dots \eta_{\xi_\nu}$ , (the interchanges, that is to say, between one  $h$  and another  $h$ , or between one  $\eta$  and another  $\eta$ , leaving  $P$  unaltered), it might be shown that the value of  $\mathfrak{S}_{v,\nu}$  resulting from the introduction of this more general value of  $R$  would (as for the particular value assumed) always be expressible as an integral function of the roots; and consequently, if  $P$  be taken of the same dimensions in the roots as the numerator of  $R$  previously assumed, that is  $v\nu$ ,  $\mathfrak{S}_{v,\nu}$  would continue to be (unless indeed it vanish) identical (to some numerical factor *près*) with the corresponding simplified residue. If, on the other hand,  $P$  be taken of less than  $v\nu$  dimensions in  $h$  and  $\eta$ , we know *a priori* that  $\mathfrak{S}_{v,\nu}$  must vanish, as otherwise we should have a conjunctive of a weight less than the minimum weight. When  $P$  is of the proper amount of weight  $v\nu$ , it is I think probable that another condition as to the *distribution* of the weight will be found to be necessary in order that  $\mathfrak{S}_{v,\nu}$  may not vanish, namely, that the highest power of any single  $h$  in  $P$  shall not exceed  $v$ , nor the highest power of any single  $\eta$  exceed  $\nu$ . But as I have not had leisure to enter upon the inquiry, the verification or disproval of this supposed law, and more generally the evolution of the allotropic numerical factor introduced into  $\mathfrak{S}_{v,\nu}$  by assigning any particular form to  $P$  satisfying the necessary conditions of amount and distribution of weight, must be reserved, amongst other points connected with the theory of the remarkable forms (19) Art. 15, as a subject for future investigation.

Art. 31. A property of continued fractions, which, if known, I have not met with in any treatise on the subject (but which has been already cursorily alluded to in these pages), gives rise to a remarkable property of reciprocity connecting  $\tau$  and  $t$  severally with  $\mathfrak{S}$  in the syzygetic equation  $\tau f - t\phi + \mathfrak{S} = 0$ .

Let the successive convergents to the ordinary continued fraction

$$\frac{1}{q_1 + \frac{1}{q_2 + \frac{1}{q_3 + \dots + \frac{1}{q_{i-1} + \frac{1}{q_i}}}}}$$

be called

$$\frac{l_1}{m_1}, \frac{l_2}{m_2}, \dots, \frac{l_{i-1}}{m_{i-1}}, \frac{l_i}{m_i},$$

respectively; it is well known that

$$m_{i-1}l_i - m_i l_{i-1} = (-)^{i-1} 1;$$

but I believe that it has not been observed that this is only the extreme case of a much more general equation, namely

$$m_{i-\rho}l_i - m_i l_{i-\rho} = (-)^{i-\rho} \mu_{\rho-1}^*,$$

where  $\mu_1, \mu_2 \dots \mu_i$  denote respectively the denominators to the convergents to the continued fractions formed with the quotients taken in a reverse order, that is, the continued fraction

$$\frac{1}{q_i + \frac{1}{q_{i-1} + \frac{1}{q_{i-2} + \dots + \frac{1}{q_2 + \frac{1}{q_1}}}}.$$

This is easily proved when  $\rho = 1$ ;  $\mu_0$  is of course (as usual) to be considered 1. So more simply for the improper continued fraction,

$$\frac{l_i}{m_i} = \frac{1}{q_1 - \frac{1}{q_2 - \dots - \frac{1}{q_{i-1} - \frac{1}{q_i}}}},$$

of which the convergents are supposed to be

$$\frac{l_1}{m_1}, \frac{l_2}{m_2} \dots \frac{l_{i-1}}{m_{i-1}}, \frac{l_i}{m_i},$$

and the reverse fraction

$$\frac{1}{q_i - \frac{1}{q_{i-1} - \dots - \frac{1}{q_2 - \frac{1}{q_1}}}},$$

of which the convergents are supposed to be

$$\frac{\lambda_1}{\mu_1}, \frac{\lambda_2}{\mu_2} \dots \frac{\lambda_i}{\mu_i},$$

we have the more simple equation

$$l_i m_{i-\rho} - l_{i-\rho} m_i + \mu_{\rho-1} = 0.$$

And it is well known, or at all events easily demonstrable, that,

$$\frac{l_{i-1}}{l_i} = \frac{1}{q_i - \frac{1}{q_{i-1} - \frac{1}{q_{i-2} - \dots - \frac{1}{q_2}}}},$$

$$\frac{m_{i-1}}{m_i} = \frac{1}{q_i - \frac{1}{q_{i-1} - \frac{1}{q_{i-2} - \dots - \frac{1}{q_2 - \frac{1}{q_1}}}}}.$$

Art. 32. If now we use subscript indices to denote the degree in  $x$  of the quantities to which they are affixed, we have the general syzygetic equation

$$K\tau_{n-t-1}f_m - Kt_{m-t-1}\phi_n + K\mathfrak{D}_t = 0,$$

where  $K$ , a constant (which I have given the means of determining in the first section), being rightly assumed,  $K\tau_{n-t-1}$ ,  $Kt_{m-t-1}$  become the numerator and denominator respectively of one of the convergents to  $\frac{\phi}{f}$ , expressed as

\* See *London and Edinburgh Philosophical Magazine*, "On a Fundamental Theorem in the Theory of Continued Fractions," Vol. vi., October, 1853. [See below.]



an improper continued fraction, and  $K\mathfrak{D}_i$  becomes the denominator to one of the convergents to  $\frac{t_{m-1}}{f}$ , or, which is the same thing, to  $\frac{\tau_{n-1}}{\phi}$  \*. Conversely, it is obvious that if we adopt as our primitive functions  $cf_m$  and  $t_{m-1}$ ,  $c$  being the value of  $K$  when  $i=0$ , we shall obtain as the general form of our syzygetic equation, bearing in mind that  $(m-1)$  now replaces  $n$ ,

$$cK'\tau_{n-i-1}f_m - K'\mathfrak{D}_{m-i-1}t_{m-1} + K't_i = 0;$$

and similarly, if we adopt as our primitive functions  $\tau_{n-1}$  and  $c\phi_n$ , we obtain for our general syzygetic equation, observing that  $(n-1)$  now replaces  $m$ ,

$$K'\mathfrak{D}_{n-i-1}\tau_{n-1} - cK't_{m-i-1}\phi_n + K'\tau_i = 0;$$

so that (making abstraction of the constant factors and looking merely to the forms of the several functions which enter into the equations) we see that on the first hypothesis, namely of  $t_{m-1}$  being substituted for  $\phi_n$ , the conjunctives of each degree in  $x$  change places with the second conjunctive factors, that is the original multipliers of  $\phi$  of the same degree in  $x$ , and *vice versa*; and in the second hypothesis, where  $\tau_{n-1}$  takes the place of  $f_m$ , the conjunctives of each degree in  $x$  change places with the first conjunctive factors, that is the original multipliers of  $f$  of the same degree in  $x$ , and *vice versa*;  $t_{m-1}$  and  $\tau_{n-1}$  being respectively multipliers of  $\phi$  and  $f$ , such that the difference of the respective products is independent of  $x$ . These results ought to be capable of being verified by aid of our general formulæ for  $t, \tau, \mathfrak{D}$ , and as this verification will serve to exhibit in a clearer light the nature of the reciprocity between the conjunctives and the conjunctive factors, it may be not uninteresting to set it out.

Art. 33. As usual, let  $h_1, h_2 \dots h_m$  be the roots of  $fx$ , and  $\eta_1, \eta_2 \dots \eta_n$  the roots of  $\phi x$ ; the last conjunctive factor to  $\phi$ , which is of the degree  $(m-1)$  in  $x$ , will be represented, neglecting powers of  $(-)$ , by  $t_{m-1}$ , where

$$t_{m-1} = \Sigma (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_{m-1}}) \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_{m-1}} \\ \eta_1, \eta_2 \dots \eta_n \end{bmatrix}}{\begin{bmatrix} h_{q_m} \\ h_{q_1}, h_{q_2} \dots h_{q_{m-1}} \end{bmatrix}}.$$

If now we for greater simplicity make  $t_{m-1} = t(x)$ , and call the roots of  $t, \eta'_1, \eta'_2 \dots \eta'_{m-1}$ , any such quantity as

$$\begin{aligned} \left[ \begin{matrix} h_{q_m} \\ \eta'_1, \eta'_2 \dots \eta'_{m-1} \end{matrix} \right] &= t(h_{q_m}) = (h_{q_m} - h_{q_1})(h_{q_m} - h_{q_2}) \dots (h_{q_m} - h_{q_{m-1}}) \\ &\quad \times \frac{\phi(h_{q_1})\phi(h_{q_2}) \dots \phi(h_{q_{m-1}})}{(h_{q_m} - h_{q_1})(h_{q_m} - h_{q_2}) \dots (h_{q_m} - h_{q_{m-1}})} \\ &= \phi(h_{q_1})\phi(h_{q_2}) \dots \phi(h_{q_{m-1}}) \\ &= R \frac{1}{\phi(h_{q_m})}, \end{aligned}$$

\* Since  $i$  is always supposed less than  $n$  ( $n$  being the degree of the lower degree of the two functions  $f$  and  $\phi$ ), the fact of the last quotient to  $\frac{t_{m-1}}{f}$  being wanting to  $\frac{\tau_{n-1}}{\phi}$  will not affect the accuracy of the statement in the text above, since this latter will contain as many quotients as can in any case be required for expressing  $\mathfrak{D}_i$ .

$R$  denoting a constant independent of the root  $h_{q_m}$  selected, in fact the resultant of the two functions  $fx$  and  $\phi x$ , that is to say,

$$\phi(h_1)\phi(h_2)\phi(h_3)\dots\phi(h_m).$$

But by our general formulæ the simplified residue to  $fx$  and  $t(x)$  of the  $i$ th degree in  $x$  will be represented by

$$\mathfrak{S}'_{i,0} = \Sigma (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_i}) \left\{ \frac{\begin{bmatrix} h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \\ \eta'_{1,} \quad \eta'_{2,} \quad \dots \eta'_{m-1} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots h_{q_i} \\ h_{q_{i+1}}, & h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}} \right\};$$

therefore

$$\begin{aligned} \mathfrak{S}'_{i,0} &= \Sigma (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_i}) \times \left\{ R^{m-i} \frac{\phi(h_{q_{i+1}})^{-1} \phi(h_{q_{i+2}})^{-1} \dots \phi(h_{q_m})^{-1}}{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots h_{q_i} \\ h_{q_{i+1}}, & h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}} \right\} \\ &= R^{m-i-1} \Sigma (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_i}) \frac{\phi(h_{q_1})\phi(h_{q_2}) \dots \phi(h_{q_i})}{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots h_{q_i} \\ h_{q_{i+1}}, & h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}}, \end{aligned}$$

or

$$\mathfrak{S}'_i = R^{m-i-1} t_i,$$

the relation which was to be obtained. So conversely, in precisely the same manner, calling  $t'_i$  the conjunctive factor of the degree  $i$  in  $x$  to  $t(x)$  in the syzygetic equation which connects  $fx$  and  $t(x)$  with a corresponding simplified residue, we have

$$\begin{aligned} t'_i &= \Sigma (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_i}) \frac{\begin{bmatrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ \eta'_{1,} \quad \eta'_{2,} \quad \dots \eta'_{m-1} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots h_{q_i} \\ h_{q_{i+1}}, & h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}} \\ &= R^{i-1} \Sigma (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_i}) \frac{\phi(h_{q_{i+1}})\phi(h_{q_{i+2}}) \dots \phi(h_{q_m})}{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots h_{q_i} \\ h_{q_{i+1}}, & h_{q_{i+2}} \dots h_{q_m} \end{bmatrix}} \\ &= R^{i-1} \mathfrak{S}_i, \end{aligned}$$

the conjugate equation to the one previously obtained\*.

And evidently the same reasoning serves to establish the reciprocity, or rather reciprocal convertibility, between the  $\mathfrak{S}$  series and the  $\tau$  series, when in lieu of the original primitives  $fx$  and  $\phi x$  we take as our primitives  $\tau(x)$  and  $\phi x$ ,  $\tau(x)$  being the function which satisfies the equation

$$\tau(x)fx - t(x)\phi x + \mathfrak{S} = 0.$$

\* M. Hermite, by a peculiar method, first discovered one of these two conjugate relations of reciprocity, applicable to the case of Sturm's theorem, where  $\phi x = f'x$ , and I am indebted to him for bringing the subject under my notice.

Art. 34. It may be remarked that if  $n = m - 1$ , the last syzygetic equation being thus  $t_{m-1}\phi_{m-1} - \tau_{m-2}f_m - \mathfrak{S}_0 = 0$ , when  $t_{m-1}$  and  $f_m$  are taken as the primitives, the corresponding equation will be of the form

$$t'_{m-1}t_{m-1} - \tau'_{m-2}f_m + \mathfrak{S}'_0 = 0;$$

these two equations must therefore be identical, and consequently  $t'_{m-1} = \phi_{m-1}$  (to a numerical factor *près*), so that  $t_{m-1}$  and  $\phi_{m-1}$  are reciprocal forms; this is also obvious from the consideration that  $t'_{m-1}$  must, by the general law of reciprocity (established above), be a residue to  $(f_m, \phi_{m-1})$ , which the latter function itself may be considered to be. Or the same thing is obvious directly, by writing

$$t_{m-1} = t(x) = \sum (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_{m-1}}) \frac{\phi(h_{q_1})\phi(h_{q_2}) \dots \phi(h_{q_{m-1}})}{(h_{q_m} - h_{q_1})(h_{q_m} - h_{q_2}) \dots (h_{q_m} - h_{q_{m-1}})},$$

and then making

$$\begin{aligned} t'_{m-1} &= \sum (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_{m-1}}) \frac{t(h_{q_1})t(h_{q_2}) \dots t(h_{q_{m-1}})}{(h_{q_m} - h_{q_1})(h_{q_m} - h_{q_2}) \dots (h_{q_m} - h_{q_{m-1}})} \\ &= R^{m-1} \sum (x - h_{q_1})(x - h_{q_2}) \dots (x - h_{q_{m-1}}) \frac{\phi(h_{q_1})^{-1} \dots \phi(h_{q_{m-1}})^{-1}}{(h_{q_m} - h_{q_1}) \dots (h_{q_m} - h_{q_{m-1}})} \\ &= R^{m-2} \sum (x - h_{q_1}) \dots (x - h_{q_{m-1}}) \frac{\phi(h_{q_m})}{(h_{q_m} - h_{q_1}) \dots (h_{q_m} - h_{q_{m-1}})}, \end{aligned}$$

or finally,

$$t'_{m-1} = R^{m-2} \phi,$$

as was to be shown.

### SECTION III.

*On the application of the Theorems in the preceding Section to the expression in terms of the roots of any primitive function of Sturm's auxiliary functions, and the other functions which connect these with the primitive function and its first differential derivative.*

Art. 35. The formulæ in the preceding Section had reference to the case of two absolutely independent functions and their respective systems of roots: when the functions become so related that the roots of the one system become explicitly or implicitly functions of the roots of the other system, the formulæ will become expressible in terms of these latter alone, and in some cases the terms (of which the sum is always essentially integral) will become separately and individually representable under an integral form. Such, as I shall proceed to show, is the case for two functions, of which one



is the differential derivative of the other. When  $f$  and  $\phi$  are thus related, so that  $\phi = \frac{df}{dx}$ , calling as before  $h_1, h_2 \dots h_m$  the roots of  $f$ , and  $\eta_1, \eta_2 \dots \eta_{m-1}$  the roots of  $\phi$ , we shall have in general

$$\begin{aligned} \left[ \begin{matrix} h_{q_{i+1}} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{matrix} \right] &= (h_{q_{i+1}} - \eta_1)(h_{q_{i+1}} - \eta_2) \dots (h_{q_{i+1}} - \eta_{m-1}) \\ &= f' h_{q_{i+1}} = \left[ \begin{matrix} h_{q_{i+1}} \\ h_{q_1}, h_{q_2} \dots h_{q_i}, h_{q_{i+2}} \dots h_{q_m} \end{matrix} \right] = \left[ \begin{matrix} h_{q_{i+1}} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{matrix} \right] \times \left[ \begin{matrix} h_{q_{i+1}} \\ h_{q_{i+2}}, h_{q_{i+3}} \dots h_{q_m} \end{matrix} \right]. \end{aligned}$$

Consequently

$$\begin{aligned} \left[ \begin{matrix} h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{matrix} \right] &= \left[ \begin{matrix} h_{q_{i+1}} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{matrix} \right] \times \left[ \begin{matrix} h_{q_{i+2}} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{matrix} \right] \times \dots \\ &\quad \times \left[ \begin{matrix} h_{q_m} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{matrix} \right] \\ &= \left[ \begin{matrix} h_{q_{i+1}} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{matrix} \right] \times \left[ \begin{matrix} h_{q_{i+1}} \\ h_{q_{i+2}}, h_{q_{i+3}} \dots h_{q_{m-1}} \end{matrix} \right] \\ &\quad \times \left[ \begin{matrix} h_{q_{i+2}} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{matrix} \right] \times \left[ \begin{matrix} h_{q_{i+2}} \\ h_{q_{i+1}}, h_{q_{i+3}} \dots h_{q_{m-1}} \end{matrix} \right] \\ &\quad \times \dots \times \left[ \begin{matrix} h_{q_m} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{matrix} \right] \times \left[ \begin{matrix} h_{q_m} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_{m-1}} \end{matrix} \right] \\ &\quad \times \frac{\left[ \begin{matrix} h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{matrix} \right]}{\left[ \begin{matrix} h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \\ h_{q_1}, h_{q_2} \dots h_{q_i} \end{matrix} \right]}. \end{aligned}$$

Hence

$$\begin{aligned} &= \left[ \begin{matrix} h_{q_{i+1}} \\ h_{q_{i+2}}, h_{q_{i+3}} \dots h_{q_m} \end{matrix} \right] \times \left[ \begin{matrix} h_{q_{i+2}} \\ h_{q_{i+1}}, h_{q_{i+3}} \dots h_{q_m} \end{matrix} \right] \times \dots \times \left[ \begin{matrix} h_{q_m} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_{m-1}} \end{matrix} \right] \\ &= (-)^{\frac{1}{2}(m-i)(m-i-1)} \zeta(h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m}), \end{aligned}$$

the  $\zeta$  denoting the operation of taking the product of the squares of the differences of the quantities which this symbol governs. Hence the Bezoutian secondary to  $f$  and  $f'$  of the  $(m-i-1)$ th degree in  $x$ , namely

$$(-)^{i \frac{i-1}{2}} \Sigma (x - h_{q_{i+1}})(x - h_{q_{i+2}}) \dots (x - h_{q_m}) \frac{\left[ \begin{matrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ \eta_1, \eta_2 \dots \eta_{m-1} \end{matrix} \right]}{\left[ \begin{matrix} h_{q_1}, h_{q_2} \dots h_{q_i} \\ h_{q_{i+1}}, h_{q_{i+2}} \dots h_{q_m} \end{matrix} \right]},$$

becomes

$$\begin{aligned} &(-)^{i(i-1)} \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_i})(x - h_{q_{i+1}})(x - h_{q_{i+2}}) \dots (x - h_{q_m}) \\ &= \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_i})(x - h_{q_{i+1}})(x - h_{q_{i+2}}) \dots (x - h_{q_m}), \end{aligned}$$

since  $(-)^{i(i-1)} = 1$ ; this gives the well-known formulæ (enunciated \* by me in the *London and Edinburgh Philosophical Magazine* for 1839) for expressing M. Sturm's auxiliary functions in terms of the roots of the primitive, and which I therein stated were immediately deducible from the general formulæ (also enunciated in the same paper) applicable to any two functions. These more general formulæ appear to have completely escaped the notice of M. Sturm and others, who have used the special formulæ applicable to the case of one function becoming the first differential derivative of the other.

Art. 36. In precisely the same manner, if we form as usual the ordinary syzygetic equation

$$tf'x - \tau fx + \mathfrak{S} = 0,$$

we may find the different values of  $t$  given by the complementary formulæ; and using  $t_i$  to denote the multiplier of the degree  $i$  in  $x$ , that is appertaining to the *residue* of the degree  $(m-i-1)$  in  $x$ , we have

$$t_i = \Sigma \frac{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots & h_{q_i} \\ \eta_1, & \eta_2 & \dots & \eta_{m-1} \end{bmatrix}}{\begin{bmatrix} h_{q_1}, & h_{q_2} & \dots & h_{q_i} \\ h_{q_{i+1}}, & h_{q_{i+2}} & \dots & h_{q_m} \end{bmatrix}} (x-h_{q_1})(x-h_{q_2}) \dots (x-h_{q_i})$$

$$= \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_i})(x-h_{q_1})(x-h_{q_2}) \dots (x-h_{q_i}).$$

Art. 37. Thus, if we make  $i = m-1$ ,

$$f_1'x = t_{m-1} = \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_{m-1}})(x-h_{q_1})(x-h_{q_2}) \dots (x-h_{q_{m-1}}).$$

It is evident from the form of  $f_1'x$  that it possesses relative to  $fx$ , the same property as  $f'x$ , I mean the property that when  $x$  is indefinitely near to a real root of  $fx$ , and is passing from the inferior to the superior side of such root,  $\frac{f_1'x}{fx}$  like  $\frac{f'x}{fx}$  will pass from being negative to being positive, or in other words,  $f_1'x$  and  $f'x$  have always the same sign in the immediate vicinity to a real root of  $fx$ . Hence it follows that  $f_1'x$  might be used instead of  $f'x$ , to produce, by the Sturmiian process of common measure, a series of auxiliary functions, which with  $fx$  and  $f_1'x$  would form a rhizoristic series, that is a series for determining (as in the manner of M. Sturm's ordinary auxiliaries) the number of real roots of  $fx$  comprised within given limits. The rhizoristic series generated by this process will, it is easily seen, be (to a constant factor *près*) the denominators (reckoning  $+1$  as the denominator in the zero place) of the successive convergents to  $\frac{f'x}{fx}$  thrown under the form

[\* p. 45 above.]

of a continued fraction  $\frac{1}{q_1 - q_2} \dots \frac{1}{q_{n-1} - q_n}$ ; M. Sturm's own rhizoristic series, on the contrary, will be (to a constant factor *près*) the denominators of the convergents to the inverse fraction  $\frac{f'_1 x}{fx}$ , which will be of the form  $K \left( \frac{1}{q_n - q_{n-1}} \dots \frac{1}{q_2 - q_1} \right)$ ; accordingly these two rhizoristic series will be equivalent as regards the number of changes and of combinations of sign (afforded by each) corresponding to any given value of  $x$ , of which of course the  $q$ 's are linear functions. This result agrees with what has been demonstrated by me\* by a more general method (in the *London and Edinburgh Philosophical Magazine*, June and July 1853), where it has been proved, by means of a very simple theorem of determinants, that the two series

$$\begin{aligned} & \frac{1}{q_1}, \frac{1}{q_1 - q_2}, \frac{1}{q_1 - q_2 - q_3}, \dots, \frac{1}{q_1 - q_2 - q_3} \dots \frac{1}{q_n}, \\ \text{and} \quad & \frac{1}{q_n}, \frac{1}{q_n - q_{n-1}}, \frac{1}{q_n - q_{n-1} - q_{n-2}}, \dots, \frac{1}{q_n - q_{n-1} - q_{n-2}} \dots \frac{1}{q_1}, \end{aligned}$$

always contain (for real values of  $q_1, q_2, q_3 \dots q_n$ ) the same number of positive and negative signs.

Art. 38. Having now determined the general values of  $\mathfrak{S}$  and  $t$  in the equation  $tf'x - \tau fx + \mathfrak{S} = 0$  as explicit integral functions of the roots of  $fx$ , the more difficult task remains to assign to  $\tau$  its value similarly expressed. This cannot readily be effected by means of substitutions in the general formulæ, the method we adopted for finding  $t$  and  $\mathfrak{S}$ ; but all the other quantities except  $\tau$  in the syzygetic equation being integral functions of the roots, it is evident that  $\tau$  also must be an integral function of the same, and to obtain it we may use the expression  $\tau = \frac{tf'x + \mathfrak{S}}{fx}$ .

To obtain the general form of  $\tau$  by direct calculation from this formula would however be found to be impracticable; the mode I adopt therefore to discover the general expression for  $\tau$  corresponding to different values of  $\mathfrak{S}$ , is to ascertain its value on the hypothesis of particular relations existing between the roots of  $fx$ , and then from the particular values of  $\tau$  thus obtained to infer demonstratively its general form, as will be seen below. The demonstration of  $\tau$  is unavoidably somewhat long,  $\tau$  being in fact represented by a double sum of partial symmetrical functions.

Using the subscript indices of each function as the syzygetic equation to denote its degree in  $x$ , we have in general

$$t_{m-i-1}f'x - \tau_{m-i-2}fx + \mathfrak{S}_i = 0,$$

[\* See below pp. 616 and 621.]

where if we make

$$h_1 - x = k_1, \quad h_2 - x = k_2 \dots h_m - x = k_m,$$

so that

$$h_i - h_\omega = k_i - k_\omega,$$

and therefore

$$\zeta(h_{\theta_1}, h_{\theta_2} \dots h_{\theta_p}) = \zeta(k_{\theta_1}, k_{\theta_2} \dots k_{\theta_p}),$$

we have in effect found

$$\mathfrak{S}_i = \sum k_{q_1} k_{q_2} \dots k_{q_i} \zeta(k_{q_{i+1}}, k_{q_{i+2}} \dots k_{q_m})$$

and

$$t_{m-i-1} = \pm \sum k_{q_1} k_{q_2} \dots k_{q_{m-i-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-i-1}});$$

we have also  $f'(x) = (-)^{m-1} \sum k_{q_1} k_{q_2} \dots k_{q_{m-1}}$ .

Let us commence with the case where  $i = 0$ ; we have then

$$\mathfrak{S}_0 = \zeta(k_1, k_2 \dots k_m),$$

$$t_{m-1} = \sum k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}});$$

we have thus

$$\begin{aligned} (-)^m \tau_{m-2} k_1 k_2 \dots k_m &= \zeta(k_1, k_2 \dots k_m) \\ &\quad - \sum k_{q_1} k_{q_2} \dots k_{q_{m-1}} \times \sum k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}). \end{aligned}$$

It may easily be verified that the negative sign interposed between the two parts of the right-hand member of the equation has been correctly taken, for

$$\zeta(k_1, k_2 \dots k_m) \text{ contains a term } k_1^{2(m-1)} k_2^{2(m-2)} \dots k_{m-2}^4 k_{m-1}^2,$$

$$\sum k_{q_1} k_{q_2} \dots k_{q_{m-1}} \text{ contains a term } k_1 k_2 \dots k_{m-2} k_{m-1},$$

and

$$\sum k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}) \text{ contains a term } k_1^{2m-3} k_2^{2m-5} \dots k_{m-2}^3 k_{m-1}^1,$$

and thus the term  $k_1^{2(m-1)} k_2^{2(m-2)} \dots k_{m-2}^4 k_{m-1}^2$ , which does not contain  $k_1 k_2 \dots k_m$ , will (as it ought to do) disappear from the right-hand side of the equation.

Now suppose

$$k_1 = k_2,$$

then

$$\zeta(k_1, k_2 \dots k_m) = 0,$$

and also

$$\zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}) = 0,$$

except when one or the other of the two disjunctive equations

$$q_1, q_2, q_3 \dots q_{m-1} = 1, 3, 4 \dots m,$$

$$q_1, q_2, q_3 \dots q_{m-1} = 2, 3, 4 \dots m,$$

is satisfied (by a disjunctive equation, meaning an equation which affirms the equality of one set of quantities with another set the same in number, each with each, but in some unassigned order).

Hence

$$\begin{aligned} & \Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}) \\ & = 2k_1 k_3 \dots k_m \zeta(k_1, k_3 \dots k_m). \end{aligned}$$

Hence when  $k_1 = k_2$ ,  $(-)^m \tau_{m-2}$  becomes

$$\frac{2}{k_1} \Sigma k_{q_1} k_{q_2} \dots k_{q_{m-1}} \zeta(k_1, k_3 \dots k_m),$$

that is

$$2\zeta(k_1, k_3 \dots k_m) \{k_1 \Sigma k_{r_3} k_{r_4} \dots k_{r_{m-1}} + 2k_3 k_4 \dots k_m\},$$

the  $\Sigma$  referring to  $r_3, r_4 \dots r_m$  supposed to be disjunctively equal to 3, 4 ...  $m$ .

Now  $\tau_{m-2}$  is of  $(m-2)$  dimensions in  $x$ , and whenever more than one equality exists between the  $k$ 's,  $\mathfrak{S}_0$  and  $t_{m-1}$  both vanish (in fact every term in each vanishes separately), and therefore  $\tau_{m-2}$ , which  $= \frac{\mathfrak{S}_0 + t_{m-1} f'x}{k_1 k_2 \dots k_m}$ , will vanish.

Hence  $(-)^m \tau_{m-2}$  must be always of the form

$$\Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_{m-1}}) \times \Psi(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}, k_{q_m}),$$

$\Psi$  denoting some integral function of  $(m-2)$  dimensions in respect of the system of quantities  $k_{q_1}, k_{q_2} \dots k_{q_m}$ . The result above obtained enables us to assign the value of

$$\Psi(k_1, k_3 \dots k_m, k_2),$$

when  $k_1 = k_2$ , namely

$$k_1 \Sigma (k_{r_3}, k_{r_4} \dots k_{r_{m-1}}) + 2k_3 k_4 \dots k_m.$$

Now for a moment suppose, selecting  $(m-1)$  terms  $k_1, k_3, k_4 \dots k_m$  out of the  $m$  terms of the  $k$  series, that

$$\begin{aligned} \Omega(k_1, k_3, k_4 \dots k_m, k_2) &= k_2^{m-2} - k_2^{m-3} S_1(k_1, k_3 \dots k_m) + k_2^{m-4} S_2(k_1, k_3 \dots k_m) \\ &\quad \pm \dots \mp k_2 S_{m-3}(k_1, k_3 \dots k_m) \pm 2S_{m-2}(k_1, k_3 \dots k_m), \end{aligned}$$

where  $S_1$  means that the quantities which it governs are to be simply added together,  $S_2$  denotes that their binary,  $S_3$  that their ternary, and in general  $S_r$  that their  $r$ -ary products are to be added together.

When  $k_1 = k_2$ ,  $\Omega$  becomes

$$\begin{aligned} & k_1^{m-2} - k_1^{m-3} \{k_1 + S_1(k_3, k_4 \dots k_m)\} + k_1^{m-4} \{k_1 S_1(k_3, k_4 \dots k_m) + S_2(k_3, k_4 \dots k_m)\} \\ & - k_1^{m-5} \{k_1 S_2(k_3, k_4 \dots k_m) + S_3(k_3, k_4 \dots k_m)\} \pm \dots \\ & \pm k_1 \{k_1 S_{m-4}(k_3, k_4 \dots k_m) + S_{m-3}(k_3, k_4 \dots k_m)\} \pm 2S_{m-2}(k_3, k_4 \dots k_m), \end{aligned}$$

which evidently equals

$$\pm \{2S_{m-2}(k_3, k_4 \dots k_m) + k_1 S_{m-3}(k_3, k_4 \dots k_m)\},$$

that is

$$\pm \{k_1 \Sigma (k_{r_3}, k_{r_4} \dots k_{r_{m-1}}) + 2k_3 k_4 \dots k_m\}.$$



Hence when  $k_1 = k_2$ ,  $\Psi = \Omega$ , and

$$(-)^m \tau_{m-2} = \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_{m-1}}) \times \Omega(k_{q_1}, k_{q_2} \dots k_{q_{m-1}}, k_{q_m});$$

and so in like manner, when  $k_1$  is equal to any one of the  $(m-1)$  quantities  $k_2, k_3 \dots k_m$ , the form of  $\tau_{m-2}$  above written will have been correctly assumed. But  $\tau_{m-2}$  may be treated as a function of  $(m-2)$  dimensions in  $k_1$ , and consequently any form of  $(m-2)$  dimensions in  $k_1$ , which fits it for  $(m-1)$  different values of  $k_1$ , must be its general form, and accordingly we have universally,

$$\begin{aligned} (-)^m \tau_{m-2} = & \Sigma \zeta(h_{q_1}, h_{q_2} \dots h_{q_{m-1}}) \times \{(x - h_{q_m})^{m-2} \\ & - (x - h_{q_m})^{m-3} S_1(x - h_{q_1}, x - h_{q_2} \dots x - h_{q_{m-1}}) \\ & + (x - h_{q_m})^{m-4} S_2(x - h_{q_1}, x - h_{q_2} \dots x - h_{q_{m-1}}) \pm \& c. \\ & \mp (x - h_{q_m}) S_{m-3}(x - h_{q_1}, x - h_{q_2} \dots x - h_{q_{m-1}}) \\ & \pm 2S_{m-2}(x - h_{q_1}, x - h_{q_2} \dots x - h_{q_{m-1}})\}. \end{aligned}$$

Art. 39. With a view to better paving our way to the general form of  $\tau$  for all values of  $i$ , let us pass over the case of  $i = 1$  and go at once to the equation

$$t_{m-3} f'x - \tau_{m-4} f'x + \mathfrak{D}_2 = 0;$$

and to better fix our ideas let  $m = 7$ , so that the equation becomes

$$t_4 f'x - \tau_3 f'x + \mathfrak{D}_2 = 0;$$

we have then, preserving the same relation as before, that is, using  $h$  to denote any root of  $fx$ , and  $k$  to denote  $h - x$ , the equation

$$\begin{aligned} \pm k_1 k_2 k_3 k_4 k_5 k_6 k_7 \tau_3 = & \Sigma k_{q_1} k_{q_2} \zeta(k_{q_3} k_{q_4} k_{q_5} k_{q_6} k_{q_7}) \\ & - \Sigma k_{q_1} k_{q_2} k_{q_3} k_{q_4} k_{q_5} k_{q_6} \times \Sigma \{k_{q_1} k_{q_2} k_{q_3} k_{q_4} \zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_4})\}; \end{aligned}$$

now  $\tau_3$  will vanish whenever more than three relations of equality exist between the  $k$ 's, for then each term in *both* of the two sums in the right-hand member of the equation above written will separately vanish; and of course three relations of equality between the same are sufficient to make all the terms in the first of these sums vanish. This relationship between the different  $k$ 's corresponding to a multiplicity 3 may arise in different ways; the multiplicity 3 may be divided into 3 units corresponding to 3 pairs of equal roots, or into 2 and 1 corresponding one set of 3 equal roots, and a second set of 2 equal roots, or may be taken *en bloc*, which corresponds to the case of one set of 4 equal roots. I shall make the first of these suppositions, which will sufficiently well answer our purpose in the case before us.

Thus I shall suppose

$$k_1 = k_4, \quad k_2 = k_5, \quad k_3 = k_6,$$

then, as above remarked,

$$\zeta(k_{q_3} k_{q_4} k_{q_5} k_{q_6} k_{q_7}) = 0$$

for all values of  $q_3, q_4, q_5, q_6, q_7$ , and therefore

$$\sum k_{q_1} k_{q_2} \zeta(k_{q_3} k_{q_4} k_{q_5} k_{q_6} k_{q_7}) = 0;$$

also  $\sum k_{q_1} k_{q_2} k_{q_3} k_{q_4} k_{q_5} k_{q_6}$  becomes

$$k_1 k_2 k_3 \{k_1 k_2 k_3 + 2k_7 (k_1 k_2 + k_1 k_3 + k_2 k_3)\},$$

and  $\zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_4})$  vanishes, except for the cases where  $q_1, q_2, q_3, q_4$  represent respectively,  $q_1$  the index 1 or 4,  $q_2$  the index 2 or 5,  $q_3$  the index 3 or 6, and  $q_4$  the index 7.

$$\text{Hence} \quad \sum k_{q_1} k_{q_2} k_{q_3} k_{q_4} \zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_4}) = 2^3 k_1 k_2 k_3 k_7 \zeta(k_1 k_2 k_3 k_7),$$

and consequently  $\tau_3$  becomes

$$\pm 8\zeta(k_1 k_2 k_3 k_7) \times \{k_1 k_2 k_3 + 2k_7 (k_1 k_2 + k_1 k_3 + k_2 k_3)\}.$$

Hence we are able to predict that the general expression for our  $\tau$  in the case before us will be

$$\begin{aligned} \tau_3 = \mp \sum \zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_7}) \\ \times \{(k_{q_4}^3 + k_{q_5}^3 + k_{q_6}^3) - (k_{q_4}^2 + k_{q_5}^2 + k_{q_6}^2)(k_{q_1} + k_{q_2} + k_{q_3} + k_{q_7}) \\ + (k_{q_4} + k_{q_5} + k_{q_6})(k_{q_1} k_{q_2} + k_{q_1} k_{q_3} + k_{q_1} k_{q_7} + k_{q_2} k_{q_3} + k_{q_2} k_{q_7} + k_{q_3} k_{q_7}) \\ - 4(k_{q_1} k_{q_2} k_{q_3} + k_{q_1} k_{q_2} k_{q_7} + k_{q_1} k_{q_3} k_{q_7} + k_{q_2} k_{q_3} k_{q_7})\}. \end{aligned}$$

For in the first place, the fact that the  $\tau$  vanishes when more than three relations of equality exist between the  $k$ 's, proves that we may assume  $\tau_3$  of the form

$$\sum \zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_7}) \times \phi(k_{q_1} k_{q_2} k_{q_3} k_{q_7}; k_{q_4} k_{q_5} k_{q_6}),$$

the semicolon (;) separating the  $k$ 's into two groups, in respect of each of which severally  $\phi$  is a symmetrical form. But if in the expression last above written for  $\tau_3$  we make

$$k_1 = k_4, \quad k_2 = k_5, \quad k_3 = k_6,$$

it becomes

$$\begin{aligned} \mp 8\zeta(k_1 k_2 k_3 k_7) \times \{(k_1^3 + k_2^3 + k_3^3) - (k_1^2 + k_2^2 + k_3^2)(k_1 + k_2 + k_3 + k_7) \\ + (k_1 + k_2 + k_3)(k_1 k_2 + k_1 k_3 + k_2 k_3 + k_1 k_7 + k_2 k_7 + k_3 k_7) \\ - 4(k_1 k_2 k_3 + k_1 k_2 k_7 + k_1 k_3 k_7 + k_2 k_3 k_7)\}. \end{aligned}$$

Now in general if

$$\sigma_r = a_1^r + a_2^r + a_3^r + \dots + a_t^r,$$

and

$$S_r = \sum (a_1 a_2 a_3 \dots a_r),$$

then

$$\sigma_r - \sigma_{r-1} S_1 + \sigma_{r-2} S_2 \pm \dots \pm r S_r = 0.$$

Consequently the sum of the terms constituting the second factor in the above expression

$$= (3-4) k_1 k_2 k_3 + (2-4) k_7 (k_1 k_2 + k_1 k_3 + k_2 k_3).$$

Hence the above expression becomes

$$\pm 8\zeta(k_1 k_2 k_3 k_7) \{k_1 k_2 k_3 + 2(k_1 k_2 + k_1 k_3 + k_2 k_3) k_7\}.$$

Thus, then, whenever  $k_1, k_2, k_3$  are respectively equal to any three of the quantities  $k_4, k_5, k_6, k_7$ , which may take place in twenty-four different ways (twenty-four being the number of permutations of four things), our  $\tau_3$  will have been correctly assumed; but  $\zeta(k_{q_1} k_{q_2} k_{q_3} k_{q_7})$  being replaceable by  $\zeta(h_{q_1} h_{q_2} h_{q_3} h_{q_7})$ , the  $\tau_3$  may be treated as a cubic function in  $k_1, k_2, k_3$ , and arranged according to the powers of  $k_1, k_2, k_3$  will contain only twenty terms; hence, since the assumed form is verified for more than twenty, that is, for twenty-four values of  $h_1, h_2, h_3$ , it follows that the assumed form is universally identical with the form of  $\tau$ , which was to be determined.

Art. 40. Now, again, in order to facilitate the conception of the general proof, let us suppose  $fx$  to be of only five dimensions in  $x, i$  still remaining 3: it will no longer be possible when we suppose a multiplicity three to prevail among the roots, to conceive this multiplicity to be distributed into three parts, for that would require the existence of three pairs of roots, there being only five. But we may, if we please, make  $h_1 = h_2 = h_3$ , and  $h_4 = h_5$ , or else  $h_1 = h_2 = h_3 = h_4$ , or in any other mode conceive the multiplicity to be divided into two parts, 2 and 1 respectively, or to be taken collectively *en bloc*. As a mode of proceeding the more remote from that last employed, I shall choose the latter supposition. Then we obtain ( $\tau$  now becoming  $\tau_{5-2-2}$ , that is  $\tau_1$ )

$$k_1 k_2 k_3 k_4 k_5 \tau_1 = \pm \Sigma k_{q_1} k_{q_2} k_{q_3} k_{q_4} \times \Sigma k_{q_1} k_{q_2} \zeta(k_{q_1} k_{q_2}),$$

and  $\zeta(k_{q_1} k_{q_2})$  will vanish, except in the case where  $q_1$  represents the indices 1 or 2 or 3 or 4, and  $q_2$  the index 5; also

$$\Sigma k_{q_1} k_{q_2} k_{q_3} k_{q_4} = k_{q_1}^4 + 4k_1^3 k_5.$$

Hence our equation becomes

$$k_1^4 k_5 \tau = \pm (k_1^4 + 4k_1^3 k_5) 4k_1 k_5 \zeta(k_1 k_5),$$

and  $\tau$  becomes

$$-4\zeta(k_1 k_5) (k_1 + 4k_5).$$

If, now, we assume for the general value of  $\tau$  in the case before us

$$\tau = \Sigma \zeta(k_{q_1} k_{q_5}) \{(k_{q_2} + k_{q_3} + k_{q_4}) - 4(k_{q_1} + k_{q_5})\},$$

when  $k_1 = k_2 = k_3 = k_4$ ,  $\tau$  becomes

$$\pm 4\zeta(k_1 k_5) \{3k_1 - (4k_1 + k_5)\},$$

that is

$$\pm 4\zeta(k_1 k_5) (k_1 + 4k_5).$$

Hence then for the two systems of values of  $h_1, h_2, h_3$ , namely

$$\left. \begin{array}{l} h_1 = h_4 \\ h_2 = h_4 \\ h_3 = h_4 \end{array} \right\} \text{ or } \left\{ \begin{array}{l} h_1 = h_5 \\ h_2 = h_5 \\ h_3 = h_5 \end{array} \right.$$

the form of  $\tau$  will have been correctly assumed. But since the derived form is a linear function of  $h_1, h_2, h_3$ , this is not enough to identify the assumed with the general form, since for such verification four systems of values must be taken, four being the number of terms in a function of three variables of the first degree. If, however, we had adopted a separation of the multiplicity three into two parts, and had started with supposing  $k_1 = k_2 = k_3, k_4 = k_5$ , we should have found that  $\tau$  would have become

$$= 6\zeta(k_1, k_5)(2k_1 + 3k_5).$$

Moreover, when these equalities subsist,

$$k_1k_2k_3k_4 + k_1k_2k_3k_5 + k_1k_2k_4k_5 + k_1k_3k_4k_5 + k_2k_3k_4k_5$$

becomes  $2k_1^3k_5 + 3k_1^2k_5^2$ , and the common factor  $k_1^2k_4$  disappears in the course of the operations for finding  $\tau$ , and eventually we have to show (in order to support the universality of the previously assumed form for  $\tau$ ) that

$$k_{q_2} + k_{q_3} + k_{q_4} - 4(k_{q_1} + k_{q_5})$$

becomes  $-2k_{q_4} - 3k_5$  when

$$k_{q_2} = k_{q_3} = k_{q_1} = k_1,$$

and

$$k_{q_4} = k_{q_5} = k_5,$$

which is evidently true. Hence then  $\tau$  will have been correctly assumed for the following cases,

$$k_1 = k_2 = k_5 = k_3$$

$$k_1 = k_2 = k_5 = k_4;$$

and also for the cases

$$\left. \begin{array}{l} k_1 = k_2 = k_3 \text{ and } k_5 = k_4 \\ k_1 = k_5 = k_3 \text{ and } k_2 = k_4 \\ k_2 = k_5 = k_3 \text{ and } k_1 = k_4 \end{array} \right\}$$

$$\left. \begin{array}{l} k_1 = k_2 = k_4 \text{ and } k_5 = k_3 \\ k_1 = k_5 = k_4 \text{ and } k_2 = k_3 \\ k_2 = k_5 = k_4 \text{ and } k_1 = k_3 \end{array} \right\},$$

that is, for eight cases in all, whereas four only would have sufficed. Hence, *ex abundantia demonstrationis*, the form assumed for  $\tau_1$  is in the case before us the general form.

Art. 41. We may now easily write down the general form which  $\tau$  assumes for all values of  $i$  and prove its correctness. If the roots be

$$h_1, h_2, h_3 \dots h_m,$$

and

$$t_{m-i-1}f'x - \tau_{m-i-2}fx + \mathfrak{S}_i = 0,$$

we shall have

$$\pm \tau_{m-i-2} = \Sigma \{ \zeta (h_{q_1} h_{q_2} h_{q_3} \dots h_{q_{m-i-1}}) \times [\sigma_{m-i-2} - \sigma_{m-i-3} S_1 + \sigma_{m-i-4} S_2 \mp \&c. \\ + (-)^{m-i-3} \sigma_1 S_{m-i-3} + (-)^{m-i-2} (\sigma_0 + 1) S_{m-i-2}] \},$$

where  $\sigma_r$  denotes in general the sum of the  $r$ th powers of the  $(i+1)$  quantities

$$(x - h_{q_{m-i}}, (x - h_{q_{m-i+1}}), \dots (x - h_{q_m}),$$

and  $S_r$  denotes in general the sum of the products of the complementary  $(m-i-1)$  quantities

$$(x - h_{q_1}), (x - h_{q_2}) \dots (x - h_{q_{m-i-1}})$$

combined  $r$  and  $r$  together. It will of course also be understood that  $\sigma_0 = i+1$ , so that  $\sigma_0 + 1 = i+2$ .

Art. 42. To prove the correctness of this general determination of the form of  $\tau_{m-i-2}$ , let us suppose in general that  $i+1$  relations of equality spring up between the  $m$  quantities  $k_1, k_2 \dots k_m$ ; we shall then easily obtain ( $N$  representing a certain numerical multiplier)

$$\pm Q = N \zeta (k_1, k_2 \dots k_{m-i-1}) \frac{\Sigma k_{q_1} k_{q_2} \dots k_{q_{m-i-1}}}{k_1^{\mu_1-1} k_2^{\mu_2-1} \dots k_{m-i-1}^{\mu_{m-i-1}-1}},$$

$k_1, k_2 \dots k_{m-i-1}$  being what the  $k$  system becomes when repetitions are excluded, and being respectively supposed to occur  $\mu_1, \mu_2 \dots \mu_{m-i-1}$  times respectively, so that

$$\mu_1 + \mu_2 + \dots + \mu_{m-i-1} = m;$$

the fractional part of the right-hand member of the equation immediately above written will be readily seen to be equivalent to

$$\Sigma \mu_{\theta_{m-i-1}} k_{\theta_1} k_{\theta_2} \dots k_{\theta_{m-i-2}}.$$

To establish the correctness of the assumed form, we must be able, as in the particular cases previously selected, to prove two things; the one, and the more difficult thing to be proved is, that when the series of distinct quantities  $k_1, k_2, k_3 \dots k_m$  become converted into  $\mu_1$  groups of  $k_1$ ;  $\mu_2$  groups of  $k_2, \dots \mu_{m-i-1}$  groups of  $k_{m-i-1}$ , then that

$$\Sigma \mu_{\theta_1} k_{\theta_2} k_{\theta_3} k_{\theta_4} \dots k_{\theta_{m-i-1}},$$

or in other terms

$$\Sigma \pm k_{\theta_1} k_{\theta_2} k_{\theta_3} \dots k_{\theta_{m-i-1}} \sum_{m-i-1}^1 (\mu_{\theta}),$$

becomes identical with

$$\sigma_{m-i-2} - \sigma_{m-i-3} S_1 \pm \&c. + (-1)^{m-i-2} (\sigma_0 + 1) S_{m-i-2}.$$

The other step to be made, and with which I shall commence, consists in showing that the number of terms in the expression last above written, considered as a function of  $(m-i-2)$ th degree of  $(i+1)$  variables, is never greater than the entire number of ways in which  $(i+1)$  quantities out of  $m$  quantities may be equated to the remaining  $(m-i-1)$  quantities, namely each of the first set respectively to all the same, or all different, or some the